These are the slides for the Java Enterprise Edition 8 for Beginners course on Udemy. They are provided free of charge to all students.

More information about the course: https://lpa.dev/u1jee8b

If you have any questions of queries, please add your feedback in the Q&A section of the course on Udemy.

Best regards,

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What is Java EE?

- Abstract Specs
- Commonly Faced Challenges – Persistence, web services, transactions, security, lose coupling etc
- *Focus is on abstract

Java implementation

- Concrete implementation – application server
- Abstract from detailed implementation

- javax.* package

App Server Examples

- Payara server (Glassfish) – https://payara.fish
- IBM OpenLiberty - https://openliberty.io
- JBOSS Wildfly - http://www.wildfly.org/

What is a Java Specification Request (JSR)?

- Formal request to Java Community Process (JCP)
- New proposal, enhancement to existing APIs eg CDI 1 -> CDI 2
- Used to group APIs into silos

What is a Reference Implementation?

- Concrete realization of the abstract JSR
- Eg JAX-RS reference implementation – Jersey
- Java EE itself is a JSR!
- Java EE 8 is JSR 366 – RI Glassfish 5

What is Jakarta EE?

- Java EE going forward
- Hosted by Eclipse Foundation
- https://jakarta.ee

Java EE/Jakarta EE vs Spring Framework

- Java EE influenced by Spring Framework
- Spring boot influenced by Java EE
- Both good platforms – use what works best and solves your problem
- No flame wars – quite pointless

Bean Discovery Annotations

- @ApplicationScoped ,
- @SessionScoped ,
- @ConversationScoped and @RequestScoped annotations
- All other normal scope types
- @Interceptor and @Decorator annotations
- All stereotype annotations (i.e. annotations annotated with
- @Stereotype),
- The @Dependent scope annotation.

The CDI Container

- Checks CDI code at boot time

- Manages component lifecycle and services
  - Create class instances
  - Add CDI features
  - Automatically destroyed

CDI Injection Points

- Field
- Constructor
- Method

- Producer
- Observer

Managed Beans

- It is not a non-static inner class.
- It is a concrete class, or is annotated with @Decorator
- It has an appropriate constructor - either:
  - The class has a constructor with no parameters, or
  - The class declares a constructor annotated with @Inject.

Bean Type of a Managed Bean

- The bean class
- Every superclass (including Object)
- All interface the class implements directly or indirectly

- Object Relational Mapping
  - ID Generation
    - Auto
    - Table
    - Identity
  - Property Access
    - Field
    - Property
  - Mapping to Table
    - @Basic
    - @Column
    - @Transient
  - Simple Java Types
  - Large Objects
    - @Lob - CLOB/BLOB
  - Enumerated Types
    - String
    - Ordinal
  - Temporal Types
  - Embeddables
  - Relationship Concepts
    - Roles
    - Directionality
    - Cardinality
    - Ordinality
    - Many-to-one
    - One-to-one
    - One-to-many
    - Many-to-many
  - Collection Mapping

- Entity Manager
  - Persistence Contexts
  - Container Managed Entity Managers
  - Transaction Management
  - Entity Manager Operations
    - Persisting
    - Finding
    - Removing
  - Cascade Operations
    - Cascade Persist
    - Cascade Remove
  - Detachment and Merging

Query Language

- JPQL
- Criteria API

- Advanced JPA
  - Advanced Queries
    - Native Queries vs JDBC
    - Defining and Executing SQL Queries
    - SQL Result Set Mapping
    - Stored Procedures
  - Entity Graphs
  - Lifecycke Events/Callbacks
  - Entity Listeners
  - Validation
  - Concurrency
  - Locking

Objects, not tables

Convenience, not ignorance

Unobtrusive, not transparent

Legacy data, new objects

Just enough, not too much

Local, but mobile

Standard API, pluggable implementation

- java.time.LocalDate
- java.time.LocalTime
- java.time.LocalDateTime
- java.time.OffsetTime
- java.time.OffsetDateTime

Enterprise JavaBeans – What are they?

EJB Development Model – Features

- Declarative Metadata
- Configuration By Exception
- Dependency Management
- Lifecycle Management
- Scalability
- Transactionality
- Security
- Portability

The EJB Architecture

- The EJB Component Model
- The EJB Container

What are Transactions?

Properties of a Transaction

- Atomicity
- Consistency
- Isolation
- Durability

General Contract – Equals Method

- It is reflexive: For any reference value x, x.equals(x) must return true.
- It is symmetric: For any reference values x and y, x.equals(y) must return true if and only if y.equals(x) return true.
- It is transitive. For any reference values x, y, and z, if x.equals(y) returns true and y.equals(z) returns true, then x.equals(z) must return true.
- It is consistent. For any reference values x and y, multiple invocations of x.equals(y) consistently return true or consistently return false, provided no information used in equals comparisons on the object is modified.

Client-server

Stateless

Cacheable

Uniform Interface

Layered System

HTTP GET METHOD - @GET

HTTP POST METHOD - @POST

HTTP PUT METHOD - @PUT

HTTP DELETE METHOD - @DELETE

- XML
- CSV
- EXCEL
- TEXT
- JSON

{“id":10,"departmentName":"IT"}

@Context

javax.ws.rs.core.Application : This class defines the components of a JAX- RS application and supplies additional metadata

javax.ws.rs.core.UriInfo : This interface provides access to the application and request URI information

javax.ws.rs.core.Request : This interface provides a method for request processing, such as reading the method type and precondition evaluation

javax.ws.rs.core.HttpHeaders : This interface provides access to the HTTP header information

javax.ws.rs.core.SecurityContext : This interface provides access to security-related information

javax.ws.rs.ext.Providers : This interface offers the runtime lookup of a provider instance, such as MessageBodyReader , MessageBodyWriter , ExceptionMapper , and ContextResolver

javax.ws.rs.ext.ContextResolver<T> : This interface supplies the requested context to the resource classes and other providers

javax.servlet.http.HttpServletRequest : This interface provides the client request information for a servlet

GET http://example.com/stuff

Accept: text/*, text/html;level=1, */*, application/xml

GET http://example.com/stuff

Accept: text/*;q=0.9, */*;q=0.1, application/json, application/xml;q=0.5

JAX-RS Filters – What Are They?

Request Filters

- Pre-Matching
- Post-Matching

Response Filters

Server Sent Events (SSE)

- HTTP PUSH API
- One way communication – Server to Client

- Event stream of text data.
- UTF-8 Encoded
- Event stream messages separated by new line “\n” characters